

CMSC 207: Chapter 2 Homework

March 2, 2026

1. Given the following premises:

$$\begin{array}{l} p \\ (p \vee r) \rightarrow s \\ \neg s \vee q \end{array}$$

Prove q .

p	given
$p \vee r$	by generalization
s	by modus ponens
$\neg s \vee q$	given
q	by elimination. ■

2. Given the following premises:

$$\begin{array}{l}\neg p \vee s \\ s \rightarrow \neg r \\ r \vee q\end{array}$$

Prove $p \rightarrow q$ using a conditional world proof.

Assume p .

$\neg p \vee s$	given
s	by elimination
$\neg r$	by modus ponens
$r \vee q$	given
q	by elimination.

Thus, $p \rightarrow q$ by conditional proof. ■